III SEMESTER	L	T	P	C
	-	-	3	1.5
20IT3L02 : JAVA PROGRAMMING LAB				

COURSE OUTCOMES:

At the end of the course student able to

CO1: Able to identify java compiler installation and JVM execution process in various environments.

CO2: Able to practice hand on experience with oops concepts in java using console and eclipse environment.

CO3: Able to simulate Basics of java programming and Exception handling.

CO4: Able to develop multi-threaded programs.

CO5: Able to develop program using JDBC connectivity to access data from database and execute different queries to get required result.

LIST OF EXPERIMENTS:

Exercise - 1 (Basics)

a) Write a JAVA program to display default value of all primitive data type of JAVA
b) Write a java program that display the roots of a quadratic equation $ax^2+bx=0$. Calculate the discriminate D and basing on value of D, describe the nature of root.
c) Five Bikers Compete in a race such that they drive at a constant speed which may or may not be the same as the other. To qualify the race, the speed of a racer must be more than the average speed of all 5 racers. Take as input the speed of each racer and print back the speed of qualifying racers.
d) Write a case study on public static void main(250 words)

Exercise - 2 (Operations, Expressions, Control-flow, Strings)

a) Write a JAVA program to search for an element in a given list of elements using binary search mechanism.
b) Write a JAVA program to sort for an element in a given list of elements using bubble sort
c) Write a JAVA program to sort for an element in a given list of elements using merge sort.
d) Write a JAVA program using StringBuffer to delete, remove character.

Exercise - 3 (Class, Objects)

a) Write a JAVA program to implement class mechanism. – Create a class, methods and invoke them inside main method.
b) Write a JAVA program to implement constructor.

Exercise - 4 (Methods)

a) Write a JAVA program to implement constructor overloading.
b) Write a JAVA program implement method overloading.

Exercise - 5 (Inheritance)

a) Write a JAVA program to implement Single Inheritance
b) Write a JAVA program to implement multi level Inheritance
c) Write a java program for abstract class to find areas of different shapes

Exercise - 6 (Inheritance, Interface)

a) Write a JAVA program give example for “super” keyword.

b) Write a JAVA program to implement Interface. What kind of Inheritance can be achieved?

Exercise – 7 (Packages)

a) Write a JAVA program illustrate class path
b) Write a case study on including in class path in your OS environment of your package.
c) Write a JAVA program that import and use the defined your package in the previous Problem

Exercise - 8 (Exception)

a) Write a JAVA program that describes exception handling mechanism
b) Write a JAVA program Illustrating Multiple catch clauses

Exercise – 9 (User defined Exception)

a) Write a JAVA program for creation of Illustrating throw
b) Write a JAVA program for creation of Illustrating finally
c) Write a JAVA program for creation of Java Built-in Exceptions
d) Write a JAVA program for creation of User Defined Exception

Exercise – 10 (Threads)

a) Write a JAVA program that creates threads by extending Thread class .First thread display "Good Morning "every 1 sec, the second thread displays "Hello "every 2 seconds and the third display "Welcome" every 3 seconds ,(Repeat the same by implementing Runnable)
b) Write a program illustrating **isAlive** and **join ()**
c) Write a Program illustrating Daemon Threads.

Exercise – 11 (Strings)

Develop a Java program that creates a string object and initializes it with your name and performs the following operations:

a) Find the length of the string object using appropriate string method.
b) Find whether the character 'a' is present in the string. If yes find the number of times 'a' appears in the name and the location where it appears.

Exercise – 12 (JDBC)

Develop a Java program using JDBC to prepare a mark list for ten students. The input is Register Number, Name, Mathematics Mark, Physics Mark and Chemistry Mark. Compute Total and Average.

TEXT BOOKS:

3. Anitha Seth, B.L.Juneja, "Java one step ahead", Oxford, 2017.
4. Herbert Schildt, "The complete Reference Java", 8th Edition, TMH, 2017

REFERENCES BOOKS:

3. Y Daniel Liang, "Introduction to Java programming", 7th Edition, Pearson, 2013.
4. Joel Murach, "Murach's Java Programming", 4th Edition, Mike Murach & Associates Inc., 2011.

E-RESOURCES:

3. https://nptel.ac.in/courses/106/105/106105191/
4. https://www.w3schools.com/java/java_data_types.asp